



Southeast University

Department of Computer Science and Engineering (CSE)

School of Sciences and Engineering

Semester: (Summer, Year: 2026)

LAB REPORT NO: 04

Course Title: Operating Systems Lab

Course Code: CSE362.2

Batch: 65

**Lab Experiment Name: Implementing loop operations (for and while loop),
array, and function in scripting.**

Student Details

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Designation : Lecturer

Submission Date : 05-08-26

Lab Report Status

Marks:

Comments:.....

Signature:.....

Date:.....

Introduction:

Bash scripting offers powerful control structures that allow users to perform repetitive tasks and manage complex data efficiently. This lab experiment focuses on extending basic shell scripting skills by introducing loop operations, array handling, and modular programming through functions. It covers iterating over numerical ranges to compute sums, storing and retrieving sequential data using arrays, structuring reusable code blocks with functions for arithmetic calculations, and utilizing nested loops to generate geometric patterns. Mastering these advanced concepts enables the development of efficient, scalable, and dynamic automation scripts for system administration.

Objectives:

- **Implement Iterative Loops:** Learn to utilize for loops to execute repetitive tasks, such as calculating the sum of specific numerical ranges.
- **Manage Data with Arrays:** Understand how to declare, populate with user input, and iterate through arrays to handle multiple data points.
- **Develop Reusable Functions:** Design and invoke user-defined functions to encapsulate mathematical operations, making code cleaner and more modular.
- **Apply Nested Control Structures:** Use nested loops to generate structured, multi-dimensional outputs, such as printing a right-angled star triangle based on a user-defined height.

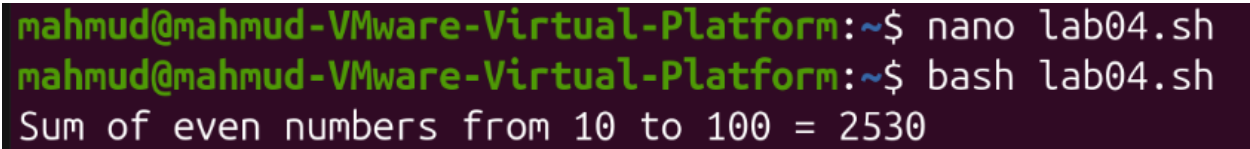
Implementation 1:

Finding the sum of even numbers between 10 and 100 using for loop

Bash code:

```
#!/bin/bash
sum=0
for ((i=10; i<=100; i+=2))
do
    sum=$((sum+i))
done
echo "Sum of even numbers from 10 to 100 = $sum"
```

Screenshot:



```
mahmud@mahmud-VMware-Virtual-Platform:~$ nano lab04.sh
mahmud@mahmud-VMware-Virtual-Platform:~$ bash lab04.sh
Sum of even numbers from 10 to 100 = 2530
```

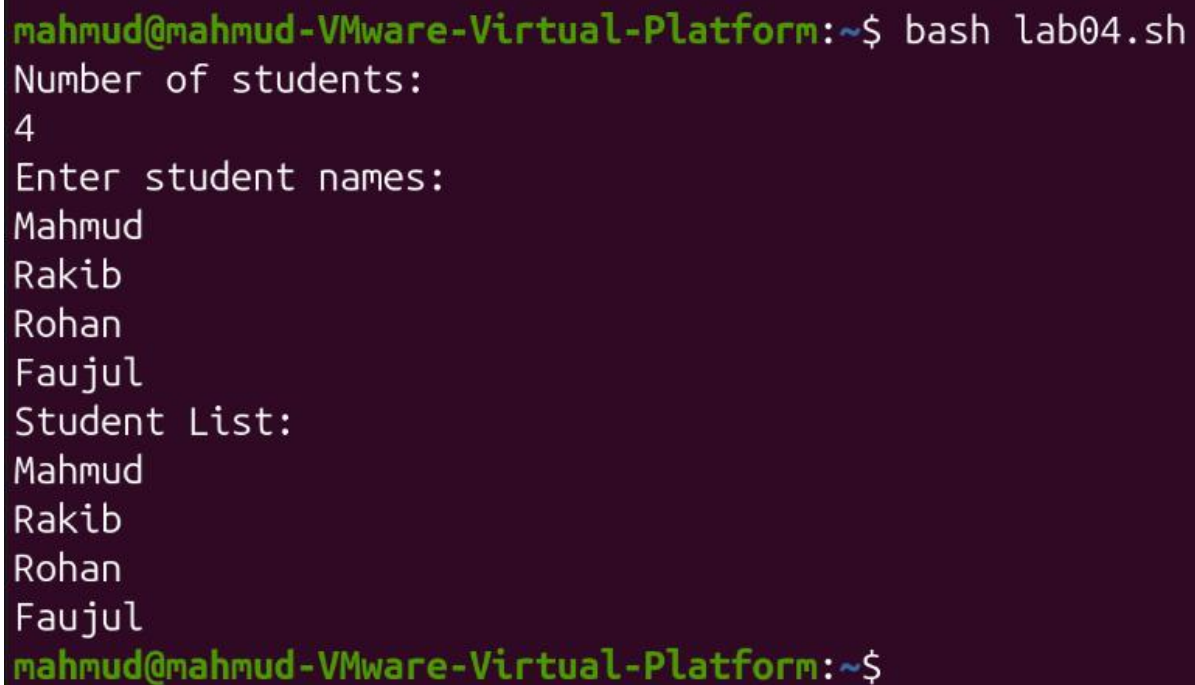
Implementation 2:

Read an array of students name using loop and then print the array

Bash code:

```
#!/bin/bash
echo "Number of students:"
read n
echo "Enter student names:"
for ((i=0; i<n; i++))
do
    read students[$i]
done
echo "Student List:"
for ((i=0; i<n; i++))
do
    echo "${students[$i]}"
done
```

Screenshot:



```
mahmud@mahmud-VMware-Virtual-Platform:~$ bash lab04.sh
Number of students:
4
Enter student names:
Mahmud
Rakib
Rohan
Faujul
Student List:
Mahmud
Rakib
Rohan
Faujul
mahmud@mahmud-VMware-Virtual-Platform:~$
```

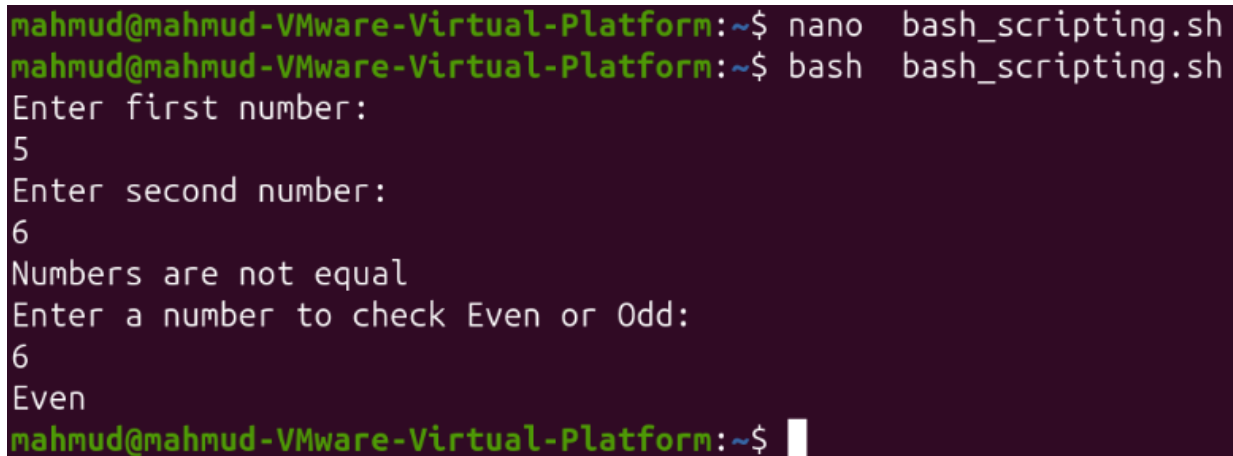
Implementation 3:

Design a function to compute addition, subtraction, multiplication, division, remainder of two different numbers. Take function parameter value as user input

Bash code:

```
#!/bin/bash
calculate()
{
    echo "Addition = $((a+b))"
    echo "Subtraction = $((a-b))"
    echo "Multiplication = $((a*b))"
    echo "Division = $((a/b))"
    echo "Remainder = $((a%b))"
}
echo "First number:"
read a
echo "Second number:"
read b
calculate
```

Screenshot:



```
mahmud@mahmud-VMware-Virtual-Platform:~$ nano  bash_scripting.sh
mahmud@mahmud-VMware-Virtual-Platform:~$ bash  bash_scripting.sh
Enter first number:
5
Enter second number:
6
Numbers are not equal
Enter a number to check Even or Odd:
6
Even
mahmud@mahmud-VMware-Virtual-Platform:~$
```

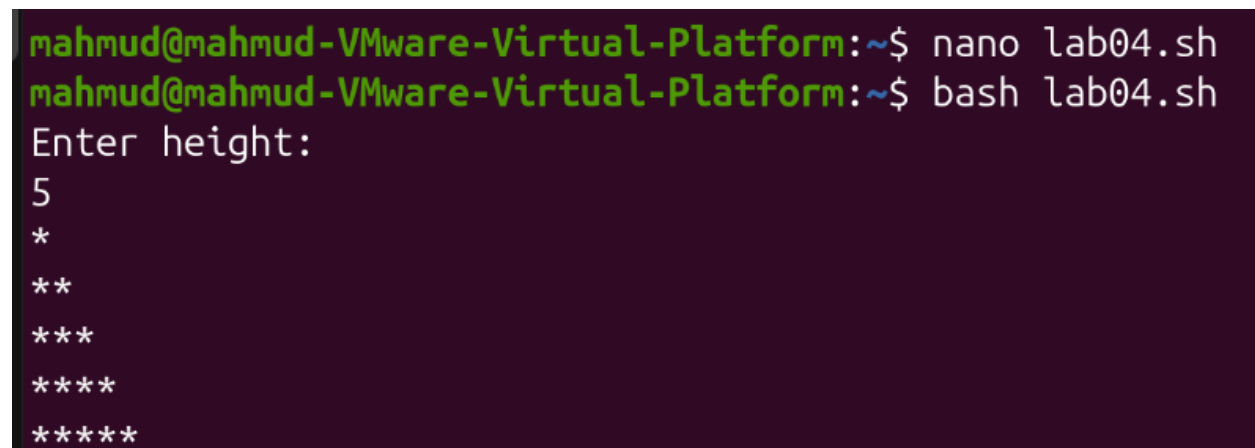
Implementation 4:

Printing the right triangle of an specific size(height of the right triangle) using loop. Take height as an user input

Bash code:

```
#!/bin/bash
echo "Enter height:"
read h
for ((i=1; i<=h; i++))
do
    for ((j=1; j<=i; j++))
    do
        echo -n "*"
    done
    echo
done
```

Screenshot:



```
mahmud@mahmud-VMware-Virtual-Platform:~$ nano lab04.sh
mahmud@mahmud-VMware-Virtual-Platform:~$ bash lab04.sh
Enter height:
5
*
**
***
****
*****
```

Conclusion:

In this experiment, advanced Bash scripting techniques were successfully implemented and executed in a Linux terminal environment. Hands-on experience was gained in controlling script execution flow using for loops, managing collections of data via arrays, and modularizing code using functions. Understanding these structural programming concepts—such as aggregating sums iteratively, processing lists of strings, and using nested loops for pattern generation—greatly enhances scripting proficiency. This provides the necessary tools to automate complex operating system processes and manage data dynamically.